

Let the Sheet Do the Work — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how each total is split, one line per mark. Every sheet extract on these papers is real data, and every value in this booklet, including each wrong value quoted in the notes, is computed by an independent Google Sheets evaluator in the generator and asserted against the hand-worked answer before typesetting. The notes under each question name the specific mistake it was built to catch. The six recurring ones on this topic: COUNT against COUNTA; quotation marks left off a criterion; a nested IF tested lowest band first; sorting one column on its own; believing a filter deletes; and believing validation cleans old data.

Paper A — Medium

30 MARKS

- 1** (a) D5 shows **150** ($=B5*C5$, 10×15) **[1]** [4]
 (b) **1015** ($180 + 220 + 240 + 150 + 225$) **[1]**
 (c) **240** — the largest *revenue*, which is the Poha row **[1]**
 (d) **11** — the five numbers in Sold add to 55, and $55 \div 5$ **[1]**
Four independent marks. Three slips to look for: a range that stops a row short, $=SUM(D2:D5)$, shows 790 and loses the Lassi; in (c) 30 is the largest price (MAX of column B), so the candidate has read the wrong column; in (d) the question asks for the average of Sold, and answering 203 (the average of Revenue) means the same wrong-column slip.
- 2** (a) **5** — only the numbers **[1]** [4]
 (b) **7** — everything except the one empty cell, so both words count **[1]**
 (c) **36** — $180 \div 5$ numbers **[1]**
 (d) **COUNT**, because it counts only cells that hold a number, so “absent” and the empty cell are not counted as sitting the test **[1]**
The COUNT/COUNTA trap in its plainest form. If (a) is 8, the candidate counted rows; 7 means COUNTA was used. For (c), 25.7 means they divided by COUNTA (the 7 non-empty cells), and 22.5 means they divided by all 8 cells; AVERAGE never does either. Award (d) only if the reason mentions numbers.
- 3** C2 (47): **Pass** **[1]** [4]
 C3 (40): **Pass** — 40 is exactly the pass mark and $\geq$ includes it **[1]**
 C4 (39): **Fail** **[1]**
 C5 (12): **Fail** **[1]**
The boundary is the point of C3. A candidate who reads $\geq$ as $>$ writes Fail for C3, and that is the only mark lost. Pass and Fail must be the words, not 1 and 0. Do not penalise missing quotation marks here: the formula is given, not written.
- 4** (a) **=MAX(C2:C8)** **[1]** [4]
 (b) **=COUNTIF(B2:B8,"A")** — range first, criterion second, quotation marks round the A **[1]**
 (c) formula **=SUMIF(B2:B8,"B",C2:C8)** — test range, criterion, then the range to add **[1]**
 (c) value **114** ($31 + 37 + 46$) **[1]**
Three traps live here. Quotation marks left off, $=COUNTIF(B2:B8,A)$, give #NAME?. The third argument left out, $=SUMIF(B2:B8,"B")$, gives 0, because Sheets adds the range it tested, which is words. $=SUM(C2:C8)$ adds both classes and gives 247. A candidate who types $=C3+C5+C7$ and gets 114 has not used the Class column: award the value mark only.

- 5 (a) highest band tested first, with its answer in quotation marks: [4]
=IF(C2>=30,"High", [1]
 second IF placed in the **false** slot, testing **C2>=20** with **"Medium"** [1]
"Low" as the last false answer, with commas and **two** closing brackets:
=IF(C2>=30,"High",IF(C2>=20,"Medium","Low")) [1]
 (b) **Medium [1]**
The order trap. =IF(C2>=20,"Medium",IF(C2>=30,"High","Low")) shows Medium for a mark of 45, and High can never appear. Award the first mark only to a formula that tests 30 before 20. In (b), Low means the candidate used > where >= was needed (a mark of exactly 20 is Medium).
- 6 (a) **=DIVIDE(A2,B2)** shows **80** (accept **=A2/B2**) [1] [2]
 (b) **=MINUS(A2,C2)** shows **600** (accept **=A2-C2**) [1]
The arguments go in the order of the sentence. =DIVIDE(B2,A2) shows 0.0125, and =MINUS(C2,A2) shows -600: the second is a negative amount of money still needed, which is visibly wrong. Either swapped version scores zero for that part.
- 7 (a) the Score column in order: **9, 14, 18, 22, 31** [1] [4]
 (a) the Name column **unchanged**: Dev, Esha, Fatima, Gopal, Hari [1]
 (b) **9** — Dev is now beside the smallest score [1]
 (c) **Fatima, Dev, Hari, Esha, Gopal** [1]
Sorting one column on its own. The scores move and the names do not, so every student now owns someone else's score: Dev (really 14) shows 9. Candidates who move names and scores together in (a) have done the correct sort, not the one described, and score 0 for (a). Part (c) is marked on its own.
- 8 (a) rows **2, 4, 6** — 35 is included by **>=** [1] [2]
 (b) it is **hidden**, not deleted: it is still in the sheet and Data > Remove filter brings it back [1]
Filter hides, it does not delete. A candidate who writes "deleted" or "moved" loses (b). The strict reading of the condition gives rows 2, 4 and loses (a); it drops the 35 in row 6. Row numbers must be given, not names. A SUM over B2:B7 still shows 191 with the filter on, because it reads the hidden rows too.
- 9 (a) **Number between 0 and 50** [1] [2]
 (b) **Reject the input** — a warning lets the bad entry in; only Reject keeps it out [1]
Computed against the rules: "Number between 0 and 50" accepts 0, 23 and 50 and refuses 51, -4 and 300. "Number greater than 0" refuses the valid mark 0 and accepts 300, so it scores nothing. Show a warning in (b) scores nothing, whatever the reason given.

Paper B — Medium

30 MARKS

- 1 **Collect 1 and Present 6** [1] [3]
Clean 2 and Organise 3 [1]
Process 4 and Analyse 5 [1]
The order that matters is Clean before Organise and Process before Analyse. The usual slip is to put Clean after Process, which is the garbage-in, garbage-out mistake: a formula run on uncleaned data gives a confident wrong number. Award each mark only if both steps in the pair are numbered correctly.
- 2 (a) (i) is **data**; (ii) is **information** [1] [2]
 (b) the data has to be **processed or analysed to answer a question** [1]
Look for the word "question" or "process/analyse". "Put it in a table" describes Organise, not information. A candidate who calls both statements data, or treats the two words as synonyms, loses (a): the syllabus wants them kept apart.

- 3** (a) **1970** ($540 + 840 + 590$, across the row) [1] [5]
 (b) **540** [1]
 (c) **370** ($840 + 180 + 90 = 1110$, $\div 3$) [1]
 (d) **470** ($590 - 120$) [1]
 (e) **60** — the smallest of all nine numbers [1]
Direction and order. In (a) 730 is the sum down the June column (B2:B4), not across row 2. In (d), swapping the cells gives -470. In (e), 120 means the candidate looked at the Aug column only (MIN of D2:D4); MIN reads all nine cells as one list. Averaging all nine instead would give 300.
- 4** (a) formula **=COUNTIF(B2:B8,"Meal")** [1] [4]
 (a) value **3** (Idli, Poha, Biryani) [1]
 (b) formula **=SUMIF(B2:B8,"Snack",C2:C8)** [1]
 (b) value **40** ($24 + 16$) [1]
Quotation marks and the third argument. =COUNTIF(B2:B8,Meal) shows #NAME?. =SUMIF(B2:B8,"Snack") shows 0: it adds the Category column, which is words. =SUM(C2:C8) shows 126 and ignores the category completely. Accept a formula with the correct logic and a different range only if it still covers rows 2 to 8.
- 5** first gap **26** [1] [3]
 second gap **16** [1]
 third gap **"None"**, with the quotation marks [1]
The gaps run from high to low, so the thresholds must be 26 then 16. Swapping them, =IF(B2>=34,"Gold",IF(B2>=16,"Silver",IF(B2>=26,"Bronze","None"))), shows Silver for a mark of 20 where Bronze is correct. Writing None without quotation marks makes the formula show #NAME? for any mark under 16. Mark each gap on its own.
- 6** (a) 47 and 36: **Pass** and **Pass** [1] [3]
 (a) 25 and 12: **Pass** and **Fail** [1]
 (b) it tests the **lowest** band first, so every mark of 20 or more is caught by the first test and Merit and Distinction can never appear; the highest band must be tested first [1]
The nested-IF order trap, to be traced rather than described. The correct formula would show Distinction, Merit, Pass, Fail for the same four marks. A candidate who writes Distinction and Merit for 47 and 36 has traced the formula they meant, not the one printed. In (b), "the brackets are wrong" or "the numbers are wrong" earns nothing: the order is the fault.
- 7** (a) **3** — rows 2, 3 and 7; capital letters do not matter, so row 3 counts [1] [4]
 (b) row 4 is **Sci**, a different word; row 6 has an **extra space** after Science, which makes it different text [1]
 (c) criteria **Dropdown** (Maths, Science, English); setting **Reject the input** [1]
 (d) it still shows **3**; the rule only checks new entries, so rows 4 and 6 must be **corrected by hand**, after which it shows **5** [1]
Validation does not clean existing data. A candidate who writes that the count is now 5 has assumed the rule fixed the old cells. Row 3 (SCIENCE) is not an error and is counted; a candidate who says row 3 is missed has not learnt that COUNTIF ignores capitals. Award (b) only if both rows are named with a reason.
- 8** (a) **=SUMIF(B2:B7,"North",C2:C7)** [1] [3]
 (b) formula **=SUMIF(B2:B7,"North",C2:C7)/COUNTIF(B2:B7,"North")** [1]
 (b) value **450** ($1350 \div 3$) [1]
The average of a group is SUMIF divided by COUNTIF. =AVERAGE(C2:C7) averages every branch and shows 350, so the zone has been ignored; dividing the North total by 6 (every row) shows 225. Both are visibly wrong against 450. Quotation marks round North are needed in both functions.

- 9 (a) a **column (or bar) chart**, because it compares categories [1] [3]
 (b) a **line chart**, because it shows change over time [1]
 (c) a **title and labels on both axes** [1]

Chart type follows the question: categories against each other get columns, change over time gets a line. A pie chart or a line chart for (a) earns nothing. For (c) both a title and axis labels are needed for the mark; "a key" alone does not earn it.

Paper C — Hard

30 MARKS

- 1 (a) **#VALUE!** [1] [5]
 (b) **4 and 5** [1]
 (c) **24** ($3 + 7 + 5 + 9$) [1]
 (d) **6** ($24 \div 4$ numbers) [1]
 (e) **10**; SUM skips the text "lost", whereas the * in C4 tries to multiply it [1]

*Functions skip text; symbols choke on it. C4 is $=B4*2$, and B4 is the word lost, so it shows #VALUE!; $=B2+B3+B4$ would show the same #VALUE!. $=SUM(B2:B4)$ gives 10 because it ignores the word. In (d), 4 means the candidate divided by all six cells, and 4.8 means they divided by COUNTA; AVERAGE divides by the 4 numbers only. In (b) the two values are 4 and 5: if the candidate swaps them, award nothing.*

- 2 (a) rows **5 and 7** [1] [5]
 (a) no other rows (both conditions must hold, and 50 is included) [1]
 (b) the first four: **Dhruv, Krish, Shaurya, Aryan** [1]
 (b) the last four: **Yuvraj, Vivaan, Atharv, Rudra** [1]
 (c) **453** — a filter hides rows but SUM still adds them [1]

Filter hides, SUM does not care. The 141 a candidate would get by adding only the two visible rows ($91 + 50$) is wrong: the formula still reads all eight rows and shows 453. Yuvraj has exactly 50, so he is on the boundary: reading "50 or more" as "more than 50" leaves only row 5. Part (b) is a whole-table sort; no two runs tie, so the order is unique. Ascending order, which is Rudra, Atharv, Vivaan ..., scores nothing.

- 3 (a) **=C2/D2*100** (or **=DIVIDE(C2,D2)*100**), which shows 160 for Aryan [1] [4]
 (b) **=COUNTIF(C2:C8,">=50")** — the operator and the number together, inside quotation marks [1]
 (c) **=SUMIF(B2:B8,"Nilgiri",C2:C8)** as the top of the fraction [1]
 (c) divided by **COUNTIF(B2:B8,"Nilgiri")**, giving **70** ($210 \div 3$) [1]

*Three precise slips. In (a) $=C2/D2$ shows 1.6 (the 100 is forgotten) and $=C2/(D2*100)$ shows 0.016; both are visibly wrong beside a strike rate of 160. In (b), $=COUNTIF(C2:C8,>=50)$ with no quotation marks shows #ERROR!, ">50" gives 3 and loses the player on exactly 50, and "50" alone counts only that one player (1). In (c), $=AVERAGE(C2:C8)$ averages every player and shows 50.6, ignoring the team.*

- 4 (a) highest band first: **=IF(A2>=12,"Danger", [1] [6]
 (a) the next two IFs in the false slots, with thresholds **9 "Alert"** and **5 "Watch"** [1]
 (a) **"Normal"** as the last false answer, all words in quotation marks, **three** closing brackets [1]
 (b) 12 is **Danger** and 9 is **Alert** [1]
 (b) 8 is **Watch** and 4 is **Normal** [1]
 (c) **Watch**, because it tests 5 first and every level of 5 or more is caught by that test, so Alert and Danger can never appear [1]**

Three tests need three IFs and three closing brackets. The formula printed in (c) shows Watch, Watch, Watch, Normal for levels of 12, 9, 8 and 4, so only the level of 4 comes out right. A candidate who writes > instead of >= shows Watch for a level of 9 where Alert is correct: mark (b) for what their formula would show. In (c), "the answer is wrong" with no reason about order earns nothing.

- 5** (a) criteria **Number between 1 and 5** [1] [4]
 (a) setting **Reject the input** [1]
 (b) only **5** is accepted (0 and 6 are outside the range; the word five is not a number) [1]
 (c) both cells stay exactly as they are, because validation checks only new entries; the owner must **correct them by hand** [1]

Validation does not clean what is already there. A candidate who says 9 and five are "rejected" or "removed" loses (c). A rule of "greater than 1" accepts 5 and 6, so it fails (a): the upper limit is missing. Show a warning in (a) scores nothing for the setting.

- 6** (a) revenues **900, 1050, 1350, 1600** [1] [6]
 (a) most units: **Chai**; most money: **Thali** [1]
 (b) **=SUM(D2:D5)**, value **4900** [1]
 (c) for money: **Thali** and **Poha** [1]
 (c) for children served: **Chai** and **Samosa** [1]
 (d) the **same data** (price and units) answers different **questions**; analysing it for money or for units gives different **information**, and so a different decision [1]

The question you ask decides the answer you get. The 225 a candidate gets from =SUM(C2:C5) is the total units, the wrong column for revenue (the right total is 4900). Neither the units ranking nor the revenue ranking has a tie, so both answers are unique. In (d) the two words must be used the right way round: data is the raw price and units, information is the answer to a question. A candidate who writes only "price is different" has the cause but not the idea: award nothing.

Paper D — Hard

30 MARKS

- 1** (a) row **7** [1] [5]
 (a) and row **8** (Jaipur has exactly 200 in July, which $\geq$ includes) [1]
 (b) the first three: **Mumbai, Kochi, Kolkata** [1]
 (b) the last four: **Delhi, Jaipur, Pune, Chennai** [1]
 (c) **245** [1]

Whole-row sorting against one-column sorting, plus a boundary. In (c) the Aug cells alone move and the names stay: Kochi is in row 5, which now holds the fourth-largest Aug value, 245, and not its own 430. A candidate who answers 430 has sorted the whole table, which is (b) not (c). In (a), reading $\geq$ as $>$ drops Jaipur and leaves only row 7. All seven Aug values are different, so the order is unique.

- 2** (a) it shows **#NAME?** [1] [5]
 (a) corrected: **=COUNTIF(B2:B6,"Drink")** [1]
 (b) it shows **0** [1]
 (b) with no third argument SUMIF adds the range it tested, column B, which is words, so there is nothing to add [1]
 (c) **3** (Chai 30, Samosa 24, Lassi 18) [1]

Quotation marks, the third argument, and a boundary. =COUNTIF(B2:B6,"Drink") shows 2; the corrected SUMIF, =SUMIF(B2:B6,"Drink",C2:C6), shows 48, which is what the student wanted. In (c), ">18" would show 2 and lose Lassi, who sold exactly 18. A candidate who explains (b) as "it should be a number" has missed that the range being added is the text column.

- 3** (a) 85 and 72 both show **B** [1] [5]
 (a) 45 shows **C** and 30 shows **D** [1]
 (b) **=IF(B2>=80,"A", the highest band tested first [1]**
 (b) **IF(B2>=60,"B",IF(B2>=40,"C","D"))** in the false slot, thresholds 60 and 40 [1]
 (b) every grade in quotation marks and **three** closing brackets:
=IF(B2>=80,"A",IF(B2>=60,"B",IF(B2>=40,"C","D"))) [1]
The nested-IF order trap, repaired not just spotted. The printed formula tests 60 before 80, so 85 shows B where A is correct; the lower marks come out right because no earlier test catches them. The correct formula gives A, B, B, C, C, D for 80, 79, 60, 59, 40 and 39: a marker can test a candidate's formula against those six. A candidate who only swaps the letters has not repaired the order.
- 4** (a) **15**, a height of 15 cm is impossible: it is a typing slip for 150 [1] [5]
 (b) **6** and **7** [1]
 (c) **130** ($780 \div 6$) [1]
 (d) **=SUMIF(B2:B9,">=100")/COUNTIF(B2:B9,">=100")**, with the operator and number inside quotation marks [1]
 (d) value **153** ($765 \div 5$) [1]
Analysis starts with distrust. The average of 130 looks plausible for a class of Grade 7 pupils, which is exactly why it is dangerous: only MIN reveals the slip. In (d), dividing the valid total by COUNTA shows 109.3, and the word and the empty cell are the reason COUNT and COUNTA disagree (6 against 7). Without quotation marks round ≥ 100 the formula shows an error.
- 5** (a) criteria **Dropdown** with the four items Red, Blue, Green, Yellow [1] [4]
 (a) setting **Reject the input** [1]
 (b) **Number between 11 and 14** [1]
 (c) **Reds** goes into the cell anyway, with a warning flag; a warning is advice, not a barrier [1]
A dropdown is the cure for spelling, because nobody can type a spelling that is not on the list. Reds is not in the list, so only Reject keeps it out. A rule of "greater than 11" with no upper limit does not earn (b). In (c) a candidate who says "it is rejected" has described the other setting.
- 6** (i) the mistake: validation does not change cells that are already filled [1] [6]
 (i) correct version: the rule only checks entries made from now on, so the 300 is still in its cell until someone changes it by hand [1]
 (i) and the cell must be corrected, for example to a valid mark, before any total or average is trusted [1]
 (ii) the mistake: a filter hides rows, it does not delete them [1]
 (ii) correct version: the Science rows are still in the sheet, hidden, and SUM still adds every row, hidden or not [1]
 (ii) so B7 still shows **140**, not **80**; the Maths-only total would need SUMIF [1]
The standing Paper D question: two wrong statements to diagnose and mend. (i) is the validation trap, the one most students believe, and (ii) is the filter trap plus the SUM-reads-hidden-rows catch. The figures are computed: =SUM(B2:B5) shows 140 with the filter on or off; the 80 comes only from the two Maths rows. The hidden Science rows hold 60 between them. Award the mistake mark only for naming what the statement gets wrong, not for restating the statement.